VIT Vidyalankar Institute of Technology
Accredited A+ by NAAC

# Consolidated Academic Administration Plan for the Course

***Artificial Intelligence (elective) Sem V – Program INFT 2026-27 Even Semester***

***Faculty - Prof. Shashikant Mahajan & Cluster Mentor- Prof. Suja Jaychandran***

**Version 09/27-09**

**Th s available in VIT –**

| VMIS (ERP) | V-Refer and V-Live | VIT Library | VAC & MOOC Courses |
|---|---|---|---|
| Institute & Department Vision and Mission | Former IA question papers and solutions (prepared by faculty) | Former IA question papers solutions - hardcopy | Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs |
| Program Educational Objectives (PEO) | MU end semester examination question papers and solutions (prepared by faculty) | MU end semester exam question paper & solutions - by faculty, hardcopy | |
| Program Specific Outcome (PSO) | Class notes and Digital Content for the subject (scanned / typed by faculty) | All text books, reference books, e -books mentioned in the syllabus & AAP | Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified |
| Program Outcome (PO) | Comprehensive question bank, EQ, GQ, PPT, Class Test papers | Technical journals and magazines for reference | |
| Departmental Knowledge Map | Academic Administration Plan & Beyond Syllabus Activity report | VIT library has many resources e:g :- IEEE, Nimbus, xplore, EBSCO etc. | Watch former lectures captured in LMS at VIT |

## 1.a Course Objectives (Write in detail – as per NBA guidelines)

| | | |
|---|---|---|
| Cognitive | What do you want students to know? | Students should understand the fundamental concepts, principles, and methodologies of Artificial Intelligence including intelligent agents, problem-solving techniques, search strategies, knowledge representation, reasoning, planning, learning, and AI applications. They should acquire knowledge of how intelligent systems perceive environments, make decisions, and solve complex real-world problems. |
| Affective | What do you want students to think / care about? | Students should appreciate the significance of Artificial Intelligence in transforming industries and society. They should develop curiosity towards intelligent systems, ethical AI practices, and emerging AI technologies while recognizing the societal impact, opportunities, and challenges associated with AI-driven solutions. |
| Behavioural | What do you want students to be able to do? | Students should be able to analyze problems, select appropriate AI techniques, apply search and reasoning methods, represent knowledge effectively, evaluate intelligent systems, and design basic AI-based solutions for real-world applications. They should also be able to communicate AI-based findings and work collaboratively in solving practical problems. |

**Advice to Students:**

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

**The v-Refer Link**

http://vidyalankarlive.com/vrefer/index.php/apps/files/?dir=/vRefer/FE/SEM%20I/202122/Engineering%20Mechanics/GM&fileid=586945 /

**Creation of microsite or teams link – (Division wise)**

| | | Team | Team Code |
|---|---|---|---|
| **Theory** | | | |
| Div1 | SM | Odd2627-TYINFT-Div1-AI-SM | 2w8o8kc |
| Div2 | SM | Odd2627-TYINFT-Div2-AI-SM | liokaqj |
| **Lab** | | | |
| A | SM | Odd2627-TYINFT-A-Batch1-AI LAB-SM | xeeaqep |
| B | SM | Odd2627-TYINFT-B-Batch1-AI LAB-SM | rjxzqxs |
| B | SM | Odd2627-TYINFT-B-Batch2-AI LAB-SM | 6jl8y1z |
| B | SM | Odd2627-TYINFT-B-Batch3-AI LAB-SM | eh59wle |
| C | SM | Odd2627-TYINFT-C-Batch1-AI LAB-SM | thg9ehe |
| C | SM | Odd2627-TYINFT-C-Batch2-AI LAB-SM | ie76k0h |
| C | SM | Odd2627-TYINFT-C-Batch3-AI LAB-SM | 74yz59x |
| C | SM | Odd2627-TYINFT-C-Batch4-AI LAB-SM | 6nin3vp |

**Collaboration Policy:**

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

## 1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

| CO No. | Statements | Related Module/s |
|---|---|---|
| CO1 | Explain the fundamental concepts, characteristics, and applications of Artificial Intelligence and intelligent systems. | 1, 2 |
| CO2 | Apply uninformed, informed, adversarial, and local search techniques to solve AI-based problems. | 3 |
| CO3 | Demonstrate knowledge representation and reasoning techniques using propositional logic, first-order logic, and uncertainty models. | 4 |
| CO4 | Analyze planning and learning approaches used in intelligent systems and AI agents. | 2, 5 |
| CO5 | Evaluate the role of Artificial Intelligence applications in solving real-world and societal problems. | 6 |
| CO6 | Design basic AI-based solutions or models for practical applications using appropriate AI techniques. | 6 |

## 1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

| | Mapped to Learning Outcomes |
|---|---|
| CO1 | LO1.1, LO1.2, LO1.3, LO1.4, LO2.1 |
| CO2 | LO3.1, LO3.2, LO3.3, LO3.4, LO3.5 |
| CO3 | LO4.1, LO4.2, LO4.3, LO4.4 |
| CO4 | LO2.2, LO2.3, LO2.4, LO5.1, LO5.2, LO5.3, LO5.4 |
| CO5 | LO6.1, LO6.2 |
| CO6 | LO6.3, LO6.4 |

## 1.d Mapping of COs with POs (As per new detailed Syllabus)

**(List of POs is available in V-refer)**

| | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 | PO 11 | PO 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Knowledge A | Analysis | Design | Investigation | Modern Tools | Society | Environment &sustainability | Ethics T | Teamwork C | Communication | Project Mgt | Lifelong learning |
| CO 1 | 2/68 | 2/68 | 1/68 | 1/68 | 2/68 | - | - | - | - | - | - | 1/68 |
| CO 2 | 2/68 | 2/68 | 2/68 | 1/68 | 2/68 | - | - | - | - | - | - | - |
| CO 3 | 2/68 | 2/68 | 2/68 | 1/68 | 2/68 | - | - | - | - | - | - | - |
| CO 4 | 2/68 | 2/68 | 2/68 | 2/68 | 2/68 | - | - | - | - | - | - | 1/68 |
| CO 5 | 1/68 | 1/68 | 1/68 | - | 1/68 | 2/68 | 1/68 | 2/68 | - | 1/68 | - | 1/68 |
| CO6 | 2/68 | 2/68 | 3/68 | 2/68 | 3/68 | 1/68 | - | 1/68 | 1/68 | 1/68 | 1/68 | 2/68 |

## 1.e Mapping of COs with PSOs (As per new detailed Syllabus)

| | PSO 1 | PSO 2 | PSO 3 |
|---|---|---|---|
| CO 1 | 5/56 | 3/56 | - |
| CO 2 | 5/56 | 5/56 | - |
| CO 3 | 4/56 | 5/56 | - |
| CO 4 | 4/56 | 5/56 | - |
| CO 5 | 3/56 | 3/56 | 2/56 |
| CO6 | 4/56 | 5/56 | 3/56 |

## 1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

| Verticals | BSC/ESC | Program Courses | Multidisciplinary Courses | Skill Courses | HSSM | Experiential Learning | Liberal Learning |
|---|---|---|---|---|---|---|---|
| Tick suitable category | | √ | | √ | | | |

| Subject Code | Subject Name | Teaching Scheme | | | Credits Assigned | | | |
|---|---|---|---|---|---|---|---|---|
| | | Theory | Practical | Tutorial | Theory | TW/Practical | Tutorial | Total |
| PEIT21T | Artificial Intelligence | 02 | - | | 02 | - | | 02 |
| PEIT21P | Artificial Intelligence Lab | - | 02 | | - | | | 01 |

| Subject Code | Subject Name | | | | Examination Scheme | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Theory | | | Total (Theory) | Practical | | | Total (Practical) |
| | | ISA | MSE | ESE | | ISA | ESE | ORAL | |
| PEIT21T | Artificial Intelligence | 15 | 20 | 40 | 75 | - | - | - | - |
| PEIT21P | Artificial Intelligence Lab | - | - | - | - | 25 | 25 | - | 50 |

| Subject Code | Subject Name | MSE-1* | | |
|---|---|---|---|---|
| | | Q, No | Module wise % Distribution | Relevant to Bloom Taxonomy |
| PEIT21T | Artificial Intelligence | 1. | M1: 45%-50%<br>M2: 50%-55% | 2-40%, 3-20%, 4-40% |
| | | 2. | | |

| Subject Code | Subject Name | MSE-2* | | |
|---|---|---|---|---|
| | | Q, No | Module wise % Distribution | Relevant to Bloom Taxonomy |
| PEIT21T | Artificial Intelligence | 1 | M3: 50%<br>M4: 50% | 2-13%, 3-33%, 4-54% |
| | | 2 | | |

| Subject Code | Subject Name | ESE# | | |
|---|---|---|---|---|
| | | Q, No | Module wise % Distribution | Relevant to Bloom Taxonomy |
| PEIT21T | Artificial Intelligence | 1 to 4 | M1-%, M2-%, M3-%, M4-%, M5-%,M6-% | 2-20%, 3-36%, 4-38%, 5-3%, 6-3% |

*** Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

**# Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

## 1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

| Divisions | Lecture (Hrs.) | Practical (Hrs.) | | | | | Tutorial (Hrs.) | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Batch | Batch | Batch | Batch | Batch | Batch 1 | Batch 2 | Batch 3 | Batch 4 |
| Div 1 (A+B) | 02 | 02 (A1) SM | 02 (A2) RSR | 02 (A3) RSR | 02 (B1) SM | 02(B2) SM | - | - | - | - |
| Div2 (B+C) | 02 | 02 (B3) SM | 02 (C1) SM | 02 (C2) SM | 02 (C3) SM | 02 (C4) SM | - | - | - | - |

## 1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

| Division | Day | Time (at least 1 Hr. / Division) | Venue (Office Room No.) |
|---|---|---|---|
| Div 1 (A+B) | Friday | 3:45 pm -4:45 pm | INFT Staff Room |
| Div2 (B+C) | Monday | 3:45 pm -4:45 pm | INFT Staff Room |

Note: Office hours should be after academic hours of students.

## 2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

| Module No. | Module Title and Brief Details | Teaching Hrs. for each module | % Weightage in autonomous syllabus Question Papers | | | Performance Indicator Mapping |
|---|---|---|---|---|---|---|
| | | | ISA | MSE | ESE | |
| 1 | **Introduction to Artificial Intelligence:** Introduction to Artificial Intelligence (AI), history and evolution of AI, characteristics and foundations of AI, intelligent systems, categories of AI (Narrow AI, General AI, Super AI), applications of AI in healthcare, education, business, cybersecurity, and smart systems. Overview of emerging trends including Generative AI and Large Language Models. | 4 | 15-20 (Open Book Test / Quiz) | 45-50 | 10 | PI1.1.1, PI1.1.2, PI1.2.1, PI1.2.2, PI1.3.1, PI1.3.2, PI1.4.1, PI1.4.2 |
| Learning Outcome-1.1 | Explain the evolution, foundations, and characteristics of Artificial Intelligence. | | | | | |
| Learning Outcome-1.2 | Describe intelligent systems and categories of AI. | | | | | |
| Learning Outcome-1.3 | Identify AI applications in various domains. | | | | | |
| Learning Outcome-1.4 | Explain recent developments in AI including Generative AI. | | | | | |
| 2 | Intelligent Agents and Multi-Agent Systems: Agents and environments, rational agents, structure of intelligent agents, types of agents (Simple Reflex, Model-Based, Goal-Based, Utility-Based, | 5 | 20-25 (Assignments/Open Book Test / Quiz) | 50-55 | 15 | PI1.2.1, PI1.2.2, PI4.3.1, PI4.3.2 |

| Module No. | Module Title and Brief Details | Teaching Hrs. for each module | % Weightage in autonomous syllabus Question Papers | | | Performance Indicator Mapping |
|---|---|---|---|---|---|---|
| | | | ISA | MSE | ESE | |
| | Learning Agents), PEAS representation, agent communication, negotiation and bargaining, argumentation among agents, trust and reputation in multi-agent systems. | | | | | |
| Learning Outcome-2.1 | Explain agents, environments, and rational agents. | | | | | |
| Learning Outcome-2.2 | Apply PEAS representation for modelling intelligent systems. | | | | | |
| Learning Outcome-2.3 | Analyze different types and architectures of intelligent agents. | | | | | |
| Learning Outcome-2.4 | Analyze communication, negotiation, bargaining, and trust mechanisms in multi-agent systems. | | | | | |
| 3 | Problem Solving and Search Techniques:<br>AI problem-solving approach, state-space representation, uninformed search techniques including Breadth First Search (BFS), Depth First Search (DFS), Depth-Limited Search, Iterative Deepening Search. Informed search techniques including Greedy Best First Search, A* Search, and heuristic search. Adversarial search, game playing, Minimax algorithm, Alpha-Beta pruning. Local search algorithms including Hill Climbing and Genetic Algorithms. | 6 | 20 | 50 | 20 | PI2.1.1, PI2.1.2, PI2.2.1, PI2.2.2, PI2.3.1, PI2.3.2, PI2.4.1, PI2.4.2 |
| Learning Outcome-3.1 | Apply uninformed search techniques to AI problems. | | | | | |
| Learning Outcome-3.2 | Apply informed search techniques using heuristic methods. | | | | | |
| Learning Outcome-3.3 | Analyze game-playing strategies using Minimax and Alpha-Beta pruning. | | | | | |
| Learning Outcome-3.4 | Evaluate the effectiveness of different search strategies. | | | | | |
| Learning Outcome-3.5 | Apply local search and optimization techniques such as Hill Climbing and Genetic Algorithms. | | | | | |
| 4 | Knowledge Representation and Reasoning:<br>Knowledge representation in AI, propositional logic, first-order logic (syntax and semantics), forward chaining, backward chaining, unification, resolution, reasoning under uncertainty, uncertain knowledge representation, belief networks, semantics and inference in probabilistic | 6 | 20 | 50 | 20 | PI3.1.1, PI3.1.2, PI3.2.1, PI3.2.2, PI3.3.1, PI3.3.2, PI3.4.1, PI3.4.2 |

| Module No. | Module Title and Brief Details | Teaching Hrs. for each module | % Weightage in autonomous syllabus Question Papers | | | Performance Indicator Mapping |
|---|---|---|---|---|---|---|
| | | | ISA | MSE | ESE | |
| | domains. | | | | | |
| Learning Outcome-4.1 | Explain knowledge representation techniques used in AI. | | | | | |
| Learning Outcome-4.2 | Apply propositional and first-order logic for knowledge representation. | | | | | |
| Learning Outcome-4.3 | Apply forward chaining, backward chaining, unification, and resolution techniques for logical reasoning. | | | | | |
| Learning Outcome-4.4 | Analyze uncertainty handling using belief networks. | | | | | |
| 5 | Planning and Learning:<br>Planning problems in AI, state-space planning, partial-order planning, hierarchical planning, conditional planning. Learning in intelligent systems, forms of learning, theory of learning, Probably Approximately Correct (PAC) learning, statistical learning, role of learning in intelligent agent behavior. | 5 | 20 (Research Paper Review / Seminar on Planning and Learning Models) | - | 20 | PI4.1.1, PI4.1.2, PI4.2.1, PI4.2.2, PI4.3.1, PI4.3.2, PI4.4.1, PI4.4.2 |
| Learning Outcome-5.1 | Explain planning concepts and approaches used in AI. | | | | | |
| Learning Outcome-5.2 | Analyze various planning techniques. | | | | | |
| Learning Outcome-5.3 | Explain different forms of learning and learning theories used in intelligent systems. | | | | | |
| Learning Outcome-5.4 | Interpret PAC learning and statistical learning approaches. | | | | | |
| 6 | Artificial Intelligence Applications and Ethics:<br>Applications of AI in healthcare, education, business, cybersecurity, finance, smart cities, and autonomous systems. AI for societal applications and sustainable development. Ethical issues in AI including fairness, bias, accountability, transparency, privacy, explainability, and responsible AI deployment. | 4 | 20 (Presentation / Poster / Activity/Case Study Analysis (AI Application & Ethics)) | - | 15 | PI5.1.1, PI5.1.2, PI5.2.1, PI5.2.2, PI5.3.1, PI5.3.2, PI5.4.1, PI5.4.2, PI6.1.1, PI6.1.2, PI6.4.1, PI6.4.2 |
| Learning Outcome-6.1 | Analyze AI applications across various domains. | | | | | |
| Learning Outcome-6.2 | Evaluate ethical and societal implications of AI systems. | | | | | |

| Module No. | Module Title and Brief Details | Teaching Hrs. for each module | % Weightage in autonomous syllabus Question Papers | | | Performance Indicator Mapping |
|---|---|---|---|---|---|---|
| | | | ISA | MSE | ESE | |
| Learning Outcome-6.3 | Design AI-based solutions for real-world problems. | | | | | |
| Learning Outcome-6.4 | Propose AI solutions for societal and community challenges. | | | | | |
| * Insert rows for more modules in the Course **Total** | | **30** | **120** | **200** | **100** | **24** |

Note: - As an attachment Annexure is required for assessment criteria of learning outcomes.

## 2.b Prerequisite Courses

| No. | Semester | Name of the Course | Topic/s |
|---|---|---|---|
| 1 | I | Structured Programming | Programming Fundamentals, Control Structures |
| 2 | III | Data Structures and Algorithms | Searching, Trees, Graphs, Algorithm Design |
| 3 | IV | Fundamentals of Logic Circuits | Logic, Boolean Algebra, Problem Solving |

## 2.c Relevance to Future Courses

| No. | Semester | Name of the Course |
|---|---|---|
| 1 | VI | Soft Computing |
| 2 | VII | Probabilistic Graphical Models |

## 2.d

**See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)**

| Real Life Scenario | Concept Used |
|---|---|
| Google Maps Route Planning and Navigation | Search Algorithms (A*, Heuristic Search) |
| Chess, Tic-Tac-Toe and Strategy Games | Minimax, Alpha-Beta Pruning |
| Virtual Assistants (Siri, Alexa, Google Assistant) | Intelligent Agents, NLP |
| ChatGPT and Generative AI Applications | Generative AI, Learning Systems |
| Medical Diagnosis Support Systems | Knowledge Representation, Reasoning, Belief Networks |
| Cybersecurity Threat Detection | AI-based Decision Making and Learning |
| Recommendation Systems (Netflix, Amazon, YouTube) | Learning and Intelligent Systems |

| | |
|---|---|
| Autonomous Vehicles and Smart Traffic Systems | Intelligent Agents and Planning |
| Customer Support Chatbots | Knowledge Representation and NLP |
| Smart Healthcare Monitoring Systems | AI Applications and Ethical Decision Making |

## 3 Past Results – Division-Wise

| Details | Target – DEC 2026 | Dec 2025 | DEC 2024 | MAY 2024 |
|---|---|---|---|---|
| Course Passing % – Average of 2 Divisions | 100 | 99.5 | 97.75 | - |
| Marks Obtained by Course Topper (mark/75) | 75 | 95.2% | 97% | - |

| | Division A | | Division B | | Division C | |
|---|---|---|---|---|---|---|
| Year | Initials of Teacher | % Result | Initials of Teacher | % Result | Initials of Teacher | % Result |
| Dec 2025 | SM | 100 | SM | 99.5 | SM | 100 |
| Dec 2024 | RSR | 97.75 | - | - | - | - |
| Dec 2023 | - | - | - | - | - | - |

## 4 All the Learning Resources – Books and E-Resources

### 4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

| Sr. No | Textbook Titles | Author/s | Publisher | Edition | Module Nos. | Available in our Library |
|---|---|---|---|---|---|---|
| 1 | Artificial Intelligence: A Modern Approach | Stuart J. Russell, Peter Norvig | Pearson | 4th | 1–6 | Yes |
| 2 | Artificial Intelligence | Elaine Rich, Kevin Knight, Shivashankar B. Nair | McGraw Hill | 3rd | 1–6 | Yes |
| 3 | Artificial Intelligence | George F. Luger | Pearson | 6th | 1–6 | Yes |

### 4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

| Sr. No | Reference Book Titles | Author/s | Publisher | Edition | Module Nos. | Available in our Library |
|---|---|---|---|---|---|---|
| 1 | Artificial Intelligence | Patrick Henry Winston | Addison-Wesley | Artificial Intelligence | All | Yes |
| 2 | Artificial Intelligence: A New Synthesis | Nils J. Nilsson | Morgan Kaufmann | Artificial Intelligence: A New Synthesis | All | Yes |
| 3 | Computational Intelligence: A Logical Approach | David Poole, Alan Mackworth | Oxford University Press | Computational Intelligence: A Logical Approach | All | Yes |

| | | | | | | |
|---|---|---|---|---|---|---|
| 4 | Deep Learning | Ian Goodfellow, Yoshua Bengio, Aaron Courville | MIT Press | Deep Learning | All | Yes |

## 4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

| Sr. No | E- Book Titles | Author/s | Publisher | Edition | Module Nos. | Available in our Library |
|---|---|---|---|---|---|---|
| 1 | Artificial Intelligence: A Modern Approach Companion Resources | Stuart Russell, Peter Norvig | Pearson | 4th | M1–M6 | |
| 2 | Computational Intelligence: A Logical Approach | David Poole, Alan Mackworth | Oxford University Press | Online Edition | M2–M5 | |
| 3 | Introduction to Artificial Intelligence (NPTEL Course Notes) | Prof. Mausam | IIT Delhi | Latest | M1–M6 | |
| 4 | Artificial Intelligence: A Modern Approach Companion Resources | Stuart Russell, Peter Norvig | Pearson | 4th | M1–M6 | |

## 4.d Reading latest / top rated research papers (at least 5 papers)

| Name of Paper | Name of Authors (Background) | Published in | | Problem Statement | Available in our Library |
|---|---|---|---|---|---|
| | | Date | Journal | | |
| Agentic Artificial Intelligence (AI): Architectures, Taxonomies, and Evaluation of Large Language Model Agents | Arunkumar V., Gangadharan G. R., Rajkumar Buyya | 2026 | arXiv | Explores architectures, taxonomy, planning, tool use, collaboration, and evaluation of autonomous AI agents built using LLMs. (arXiv) | No |
| Agentic AI: A Comprehensive Survey of Architectures, Applications, and Future Directions | M. Abou Ali et al. | 2025 | Artificial Intelligence Review (Springer) | Provides a comprehensive study of Agentic AI architectures, applications, evaluation methods, and future research directions. (Springer Nature) | No |
| ReAct: Synergizing Reasoning and Acting in Language Models | Shunyu Yao et al. | 2023 | ICLR | Studies how AI systems can combine reasoning and action to solve complex tasks. | No |
| Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks | Patrick Lewis et al. | 2020 | NeurIPS | Investigates how external knowledge retrieval can improve factual correctness of language models. | Yes |

| | | | | | |
|---|---|---|---|---|---|
| Attention Is All You Need | Ashish Vaswani et al. | 2017 | NeurIPS | Introduces the Transformer architecture that became the foundation of modern AI and Large Language Models. | Yes |

## 4.e Based on research paper an identify the current Problem statement

| Problem Statement | | | Used in | | | |
|---|---|---|---|---|---|---|
| | Quiz | Assignment | *Mini Project* | Poster Presentation | Test | Any Other |
| How can Agentic AI systems autonomously reason, plan, collaborate, and safely execute complex real-world tasks while ensuring transparency, reliability, ethical compliance, and human oversight? | ✓ | | ✓ | ✓ | ✓ | |
| How can Large Language Model (LLM)-based AI agents integrate reasoning, external knowledge retrieval, planning, and tool usage to solve real-world problems accurately and responsibly? | ✓ | | ✓ | ✓ | ✓ | |

## 4.f Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements

| Name of the Company | To be / Contacted for | | |
|---|---|---|---|
| | Student Internship | Student Final Placement | Faculty Internship |
| Google DeepMind | ✓ | ✓ | ✓ |
| Microsoft AI | ✓ | ✓ | ✓ |
| OpenAI | ✓ | ✓ | ✓ |
| NVIDIA | ✓ | ✓ | ✓ |
| Amazon AWS AI | ✓ | ✓ | ✓ |
| IBM Watson | ✓ | ✓ | ✓ |
| TCS AI.Cloud | ✓ | ✓ | ✓ |
| Infosys Topaz | ✓ | ✓ | ✓ |
| Accenture AI | ✓ | ✓ | ✓ |
| Pratit Technologies | ✓ | ✓ | ✓ |

## 4.g Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)

| Name of the Identified Guest Speaker | Designation | Name of the Company |
|---|---|---|
| Mr. Pratap Sanap | Head – Digital Transformation & AI | Pratit Technologies, Pune |
| Mr. Anand Varade | Business Leader & CXO | IT Industry Veteran |
| Ms. Priti Munshi | Head of Product Management (BFSI) | BFSI Sector |
| Mr. Sunilkumar Shinde | Vice President | Bharti AXA Life Insurance |
| Dr. Mausam | Professor, AI Researcher | IIT Delhi |
| Dr. Debasis Samanta | Professor | IIT Kharagpur |

## 4.h Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]

| Name of the Relevant Technical Competition Identified to participate | Organized by | Date of the Event |
|---|---|---|
| Smart India Hackathon | Ministry of Education | Annual |
| Kaggle AI Competitions | Kaggle | Throughout Year |
| Google Solution Challenge | Google Developer Program | Annual |
| IEEE AI Hackathon | IEEE | Annual |
| Amazon ML Challenge | Amazon | Annual |
| Data Science Olympiad | Various Organizations | Annual |

## 4.i Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication

| University | Name of the Course | Name of Faculty | Type of Collaboration | | |
|---|---|---|---|---|---|
| | | | Exchange of Lecture Material | Joint Publication/ Research | Other |
| IIT Delhi | Artificial Intelligence | Prof. Mausam | ✓ | - | ✓ |
| IIT Kharagpur | Artificial Intelligence | Prof. Debasis Samanta | ✓ | - | ✓ |
| Stanford University | Artificial Intelligence | Andrew Ng (Course Resources) | ✓ | - | ✓ |
| UC Berkeley | Artificial Intelligence | Stuart Russell | ✓ | - | ✓ |

## 4.j Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary

| Module No. | Title of the Module | Textbook | Mention the Tile | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | Reference Book | E-books | Journal | E-Journal | Available in our Library | V-refer |
| 1 | M1 | Introduction to AI | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| 2 | M2 | Intelligent Agents | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| 3 | M3 | Search Techniques | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| 4 | M4 | Knowledge Representation | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| 5 | M5 | Planning and Learning | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| 6 | M6 | Applications and Ethics | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

## 4.k Referred to any top-rated university in that subject for content

| University | Name of the Course | Name of Faculty | Date of Delivery of the Course | Remarks |
|---|---|---|---|---|
| IIT Delhi | Introduction to Artificial Intelligence | Prof. Mausam | NPTEL Course | IIT Delhi |
| UC Berkeley | CS188 Artificial Intelligence | Faculty Team | Open Course Resources | UC Berkeley |
| Stanford University | CS221 Artificial Intelligence | Stanford AI Faculty | Open Course Resources | Stanford University |

## 4.l Faculty received any certification related to this subject. List of Certifications Identified / Done

| Course | Certifying Agency | No. of Hours | Level of the Course | | Certification | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | Introductory | Advance Skill Development | Done on | Proposed to be on | |
| Soft Computing | NPTEL | 12 Weeks | ✓ | ✓ | Apr 2026 | - | Completed |
| Quantum Computing | NPTEL | 4 Weeks | ✓ | ✓ | March 2026 | - | Completed |
| Artificial Intelligence | NPTEL | 12 Weeks | ✓ | ✓ | Oct-Nov 2025 | - | Completed |

## 4.m Completed subject wise/cluster wise training with cluster mentor. List of relevant Refresher Course Identified / Done

| Course | Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU) | Certification | | Remarks |
|---|---|---|---|---|
| | | Done on | Proposed to be on | |
| Pedagogy | Workshop by Cluster Mentor | 22 July 2026 | - | Outcome-Based Education and Active Learning |
| PBL | Workshop by Industry Mentor | 23 July 2026 | - | AI-Based Problem-Based Learning |
| Sub. Content Training | NPTEL - Artificial Intelligence: Concepts and Techniques | Oct 2025 | - | Generative AI and LLMs<br>Agentic AI Systems<br>AI Ethics and Responsible AI |

## 4.n Best Practices Identified and adopted

| No. | Item | Best Practices Identified | | |
|---|---|---|---|---|
| | | Name of Univ. 1 | Name of Univ. 2 | Name of Univ. 3 |
| 1 | Microsite | https://ocw.mit.edu/course s/electrical-engineering- and-computer-science/6- | https://ai.stanford.edu/~latombe/cs121/2011/schedule.htm | https://www.cs.utexas.edu/~ mooney/cs343/<br><br>Raymond J. Mooney |

| | | | | |
|---|---|---|---|---|
| | | 034-artificial-intelligence-fall-2010/ | | |
| 2 | Video Lectures | Prof. Patrick Henry Winston | https://online.stanford.edu/artificial-intelligence/free-content?category=All&course=All | |
| 3 | Assignments | https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-034-artificial-intelligence-fall-2010/assignments/ | https://ai.stanford.edu/~latombe/cs121/2011/schedule.htm | https://www.cs.utexas.edu/~mooney/cs343/ |
| 4 | Mini Project | -- | -- | https://www.cs.utexas.edu/~mooney/cs343/proj1/ |
| 5 | Assessment Metric | -- | -- | -- |
| 6 | Quizzes | https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-034-artificial-intelligence-fall-2010/exams/ | -- | -- |
| 7 | Labs/ Practical (PBL) | https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-034-artificial-intelligence-fall-2010/assignments/ | -- | -- |
| 8 | Tests | https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-034-artificial-intelligence-fall-2010/exams/ | -- | https://www.cs.utexas.edu/~mooney/cs343/ |
| 9 | Peer Assessment | -- | -- | -- |
| 10 | Any Other | -- | -- | -- |

## 4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

| No. | Websites / Links | Module Nos. |
|---|---|---|
| 1 | https://inst.eecs.berkeley.edu/~cs188/sp26/ | M1-M5 |
| 2 | https://nptel.ac.in/courses/106102220 | M1-M6 |
| 3 | https://www.youtube.com/playlist?list=PLoROMvodv4rOca_Ovz1DvdtWuz8BfSWL2 | M1-M6 |

## 4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

| Sr. No. | MOOC Course Link | Course conducted by – Person / University / Institute / Industry | Course Duration | Certificate (Y / N) |
|---|---|---|---|---|
| 1 | Introduction to Artificial Intelligence | NPTEL | 12 Weeks | Yes |
| 2 | AI For Everyone | Coursera | 4 Weeks | Yes |
| 3 | Machine Learning Specialization | Coursera | 3 Months | Yes |
| 4 | Generative AI for Everyone | Coursera | 4 Weeks | Yes |
| 5 | CS188 Artificial Intelligence | Berkeley | Self-paced | No |
| 6 | Elements of AI | University of Helsinki | Self-paced | Yes |

## 5 Consolidated Course Lesson Plan

| | From (date/month/year) | To (date/month/year) | Total Number of Weeks |
|---|---|---|---|
| Semester Duration | 06/07/2026 | 23/10/2026 | 15 |

| Week | Lecture no. | Module No. | Lecture Topics / MSE / BSA planned to be covered | Actual date of Completion (Handwritten) | COs Mapped | Mapping Bloom Taxonomy level | Recommended Prior Viewing / Reading | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Lecture No. (on LMS) | Chapter No./ Books/ Web Site |
| 1 | 1 | M1 | Introduction to Artificial Intelligence | | CO1 | Understand | SB, PPT | |
| | 2 | M1 | Characteristics and Foundations of AI | | CO1 | Understand | PPT + Discussion | |
| 2 | 3 | M1 | Intelligent Systems and Types of AI | | CO1 | Understand | Case Study | |
| | 4 | M1 | Applications of AI and Generative AI | | CO1 | Apply | Video + Discussion | |
| 3 | 5 | M2 | Agents and Environment | | CO1 | Understand | PPT | |
| | 6 | M2 | Rational Agents and PEAS Representation | | CO1 | Apply | Activity Based | |

| Week | Lecture no. | Module No. | Lecture Topics / MSE / BSA planned to be covered | Actual date of Completion (Handwritten) | COs Mapped | Mapping Bloom Taxonomy level | Recommended Prior Viewing / Reading | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Lecture No. (on LMS) | Chapter No./ Books/ Web Site |
| | | | | | | | Learning | |
| 4 | 7 | M2 | Structure and Types of Intelligent Agents | | CO1 | Analyze | Demonstration | |
| | 8 | M2 | Learning Agents | | CO4 | Analyze | Case Study | |
| 5 | 9 | M2 | Agent Communication, Negotiation and Bargaining | | CO4 | Analyze | Discussion | |
| | 10 | M2 | Trust and Reputation in Multi-Agent Systems | | CO4 | Analyze | Research Paper Discussion | |
| 6 | 11 | M3 | Problem Solving using AI | | CO2 | Understand | PPT | |
| | 12 | M3 | Breadth First Search (BFS) | | CO2 | Apply | Problem Solving | |
| 7 | 13 | M3 | Depth First Search (DFS), Depth Limited Search | | CO2 | Apply | Numerical Practice | |
| | 14 | M3 | Iterative Deepening Search | | CO2 | Apply | Tutorial | |
| 8 | 15 | M3 | Greedy Best First Search and Heuristic Search | | CO2 | Apply | Problem Solving | |
| | 16 | M3 | A* Search Algorithm | | CO2 | Analyze | Numerical Practice | |

| Week | Lecture no. | Module No. | Lecture Topics / MSE / BSA planned to be covered | Actual date of Completion (Handwritten) | COs Mapped | Mapping Bloom Taxonomy level | Recommended Prior Viewing / Reading | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Lecture No. (on LMS) | Chapter No./ Books/ Web Site |
| 9 | 17 | M3 | Minimax Search | | CO2 | Analyze | Game Tree Activity | |
| | 18 | M3 | Alpha-Beta Pruning | | CO2 | Analyze | Tutorial | |
| 10 | 19 | M3 | Hill Climbing and Genetic Algorithms | | CO2 | Apply | Case Study | |
| | 20 | M4 | Knowledge Representation in AI | | CO3 | Understand | PPT | |
| 11 | 21 | M4 | Propositional Logic | | CO3 | Apply | Tutorial | |
| | 22 | M4 | First Order Logic: Syntax and Semantics | | CO3 | Apply | Problem Solving | |
| 12 | 23 | M4 | Forward Chaining and Backward Chaining | | CO3 | Apply | Demonstration | |
| | 24 | M4 | Unification and Resolution | | CO3 | Analyze | Tutorial | |
| 13 | 25 | M4 | Belief Networks and Uncertain Knowledge | | CO3 | Analyze | Case Study | |
| | 26 | M5 | Planning Problem and State Space Planning | | CO4 | Understand | PPT | |
| 14 | 27 | M5 | Partial Order, Hierarchical and Conditional Planning | | CO4 | Analyze | Discussion | |
| | 28 | M5 | Forms of Learning, PAC Learning and Statistical Learning | | CO4 | Analyze | Research Paper | |

| Week | Lecture no. | Module No. | Lecture Topics / MSE / BSA planned to be covered | Actual date of Completion (Handwritten) | COs Mapped | Mapping Bloom Taxonomy level | Recommended Prior Viewing / Reading | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Lecture No. (on LMS) | Chapter No./ Books/ Web Site |
| | | | | | | | Discussion | |
| 15 | 29 | M6 | AI Applications in Healthcare, Education, Business and Cybersecurity | | CO5 | Evaluate | Case Study | |
| | 30 | M6 | AI Ethics, Fairness, Privacy and Societal Applications | | CO5, CO6 | Create | Debate + Discussion | |

## 6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

| Theory (ISA= 10M) | | | | | | | | | | | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class Participation | Assignment -1 | Assignment -2 | Assignment -3 | Assignment -4 | Assignment -5 | Assignment -6 | Assignment -7 | Assignment -8 | Assignment -9 | Assignment -10 | Assignment -11 | |
| All | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |

**Assignment marks shall be awarded only to those students who have attended the respective theory session.**

| Theory (ISA= 5M) | | | | | | | | | | | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class Participation | Open Book Test | Pop Quiz | Case Study Analysis | Presentation | Poster | Class Test | | | | | | |
| All | 02 | 01 | 01 | 01 | 01 | 01 | | | | | | 05 |

**The Open Book Test is compulsory. In addition, students must successfully complete at least three out of the remaining assessment activities to receive full marks.**

| | Practical (ISA=25) | | | | | | | | | | | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class Participation | Practical -1 | Practical -2 | Practical -3 | Practical -4 | Practical -5 | Practical -6 | Practical -7 | Practical -8 | Practical -9 | Practical -10 | Mini-Project Evaluation | Mini-Project Report | |
| All | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 05 | 25 |

**Experiment marks shall be awarded only to those students who have attended the respective Lab session.**

| Class Participation | ISA | MSE-1 | MSE-2 | ESE | Total |
|---|---|---|---|---|---|
| | | 20 | 20 | | |
| All | 15 | 20 (Average of MSE1 and MSE2) | | 40 | 75 |

## 7 Assignments / Tutorials Details

| Assignment/ Tutorial No. | Title of the Assignments / Tutorials | CO Map | Mapping Bloom Taxonomy Level | Assignment/ Tutorials given to Students on | Assignments to be submitted back on |
|---|---|---|---|---|---|
| 1 | AI Around Me – Discovering Intelligence in Everyday Life | CO1 | Understand | | |
| 2 | Engineering Intelligent Agents – Understanding AI Behaviour | CO1 | Analyze | | |
| 3 | Designing PEAS Models for Real-World Intelligent Systems | CO6 | Create | | |
| 4 | Engineering Search Strategies – Finding Intelligent Solutions | CO2 | Evaluate | | |
| 5 | Engineering Intelligent Decision Making – Game Playing, Minimax and Alpha-Beta Pruning | CO2 | Evaluate | | |
| 6 | Engineering Knowledge Representation – Representing Intelligence Effectively | CO3 | Analyze | | |

| | | | | | |
|---|---|---|---|---|---|
| 7 | Engineering Logical Reasoning – Reasoning with Knowledge | CO3 | Analyze | | |
| 8 | Engineering Intelligent Planning – Planning Actions for Goal Achievement | CO4 | Create | | |
| 9 | Engineering Learning Systems– How Intelligent Systems Improve Through Learning | CO4 | Evaluate | | |
| 10 | Engineering Responsible AI – Building Trustworthy Intelligent Systems | CO5 | Evaluate | | |
| 11 | Capstone Engineering Investigation – Engineering AI for Society | CO6 | Create. | | |

**Analysis of Assignment / Tutorial Questions and Related Resources**

| Assignment / Tutorial No. | Week No. | Type* (√) | | | Module No. | Based on # | | | Question Type (√) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | OT | CS | DTP | | Textbook | Reference Book | Other Learning Resource | Real Life Assignments | Thought Provoking |
| 1 | 2 | | √ | √ | M1 | T1, O1 | √ | √ | 2 | |
| 2 | 4 | | √ | √ | M2 | T1, R1 | √ | √ | 4 | |
| 3 | 6 | | | √ | M3 | T1, R2 | √ | √ | 6 | |
| 4 | 7 | | | √ | M3 | T1, O2 | √ | √ | 7 | |
| 5 | 8 | | √ | √ | M3 | T1, R2 | √ | √ | 8 | |
| 6 | 10 | | | √ | M4 | T1, R1 | √ | √ | 10 | |
| 7 | 11 | | √ | √ | M4 | T1, R1 | √ | √ | 11 | |
| 8 | 13 | | √ | √ | M5 | T1, R2 | √ | √ | 13 | |
| 9 | 14 | | √ | √ | M6 | T1, O3 | √ | √ | 14 | |
| 10 | 15 | √ | | √ | M5, M6 | O3 | √ | √ | 15 | √ |
| 11 | 16 | | √ | √ | M6 | T1, O3 | √ | √ | 16 | |

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)
# Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

## 8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

| Tests | Test Dates | Module No. | CO Map | MSE Question Paper Pattern | Policy |
|---|---|---|---|---|---|
| Class Test | 3rd Week | 2 | CO1, CO4 | 2 questions | Compulsory 20M |
| Pop Quiz | 13th Week | 1-5 | CO1-CO4 | 20 MCQ | Compulsory 20M |
| Open Book Test | 6th Week | 3 | CO2 | 2 Question | Compulsory 20M |
| Take Home Test | 8th Week | 4 | CO3 | 2 Questions | Compulsory 20M |
| Poster Presentation | 12th Week | 6 | CO5, CO6 | Topic wise | Compulsory |

**Failures of IA test (IA1+IA2) shall appear for IA test in the next semester. There is no provision for re-test in the same semester.**

## 9. Practical Activities

| Practical No. | Module No. | Title of the **Experiments** | Type of Experiment | | Topics to be highlighted | CO Map |
|---|---|---|---|---|---|---|
| | | | PBL | Newly Added | | |
| 1 | | Python Programming Fundamentals for AI Applications | - | Yes | Python Programming Fundamentals | CO5, CO6 |
| 2 | | Design of Intelligent Systems using PEAS Representation | - | - | Agents & Agentic Environment | CO2, CO6 |
| 3 | | Breadth First Search (BFS) | - | - | Problem Solving | CO1, CO6 |
| 4 | | Depth First Search (DFS) | - | - | Problem Solving | CO1, CO6 |
| 5 | | A* Search Algorithm | - | - | Problem Solving | CO1, CO6 |
| 6 | | Hill Climbing Search Algorithm | - | - | Problem Solving | CO1, CO5 |
| 7 | | Minimax Algorithm with Alpha-Beta Pruning | - | - | Game Playing and Problem Solving | CO2, CO5 |
| 8 | | First Order Logic and Rule-Based Reasoning | - | - | Reasoning | CO3, CO6 |
| 9 | | Forward Chaining and Backward Chaining | - | - | Reasoning | CO3, CO6 |
| 10 | | Mini Project-I: AI Problem Solver | PBL | - | Real-time Applications | CO5, CO6 |
| 11 | | Mini Project-II: AI for Social Good | PBL | - | Real-time Applications | CO4, CO5, CO6 |

## 10 Uncovering syllabus with different Activities.

| No. | Type of the Activity | Activities | Number of beneficiaries | Other Details – guest profile, feedback, mark sheet, report |
|---|---|---|---|---|
| 1 | **Experiential learning/Interaction with Outside World** | 1- Guest Lectures by Industry Expert | All | |
| | | 2- Workshops | | |
| | | 3- Mini Project | All | |
| | | 4- Industrial Visit | | |
| | | 5- Any other activity | | |
| 2 | **Collaborative & Group Activity** | 6- Poster Presentation | All | |
| | | 7- Minute Papers | All | |
| | | 8- Students Seminars | All | |
| | | 9- Students Debates | | |
| | | 10- Panel Discussion / Mock GD | | |
| | | 11- Mock Interview | | |
| | | 12- Any other activity | | |
| 3 | **Co-Curricular Activity** | 13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX) | All | |
| | | 14- Lecture Capture Usage | All | |
| | | 15- Any other activity | | |
| 4 | **Tests & Assessments** | 16- Class Tests/ Weekly Tests | All | |
| | | 17- Pop Quiz | All | |
| | | 18- Mobile App Based Quiz | | |
| | | 19- Open Book Test | All | |
| | | 20- Take Home Test | All | |

| | | | | |
|---|---|---|---|---|
| | | 21- Any other activity | | |

## 11 AAP

| No. | Programme | Course | Uploaded on V-refer | Date |
|---|---|---|---|---|
| 1 | INFT | Artificial Intelligence | Yes | 19 July 2026 |
| | | | | |
| | | | | |

## 12 Lecture Guide

| No. | Programme | Course | Uploaded on V-refer | Date |
|---|---|---|---|---|
| 1 | INFT | Artificial Intelligence | Yes | 19 July 2026 |
| | | | | |
| | | | | |

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

| | | |
|---|---|---|
| Shashikant Mahajan<br>Faculty 1 | Dr. Rasika Ransing<br>Faculty 2 | Faculty 3 |

| | | | |
|---|---|---|---|
| Mr. Laxmikant Tiwari<br>External Industry Mentor | Dr. Bharti Joshi<br>External Academic Mentor | Dr. Suja Jaychandran<br>VIT Cluster Mentor | Dr. Vidya Chitre<br>Program HOD |

## Annexure:

**Assessment Criteria of Learning Outcomes:**

| **Learning Outcomes:**<br>The Learner will: | **Assessment Criteria:**<br>The Learner can: | **Evaluated under ISA/MSE/ESE/LAB** |
|---|---|---|
| M1-LO1: Explain the evolution, foundations, and characteristics of AI. | Describe AI history, foundations, and characteristics with suitable examples. | ISA, MSE, ESE |
| M1-LO2: Differentiate categories of AI and intelligent systems. | Classify AI systems as Narrow AI, General AI, and Super AI. | ISA, MSE, ESE |
| M1-LO3: Analyze applications and emerging trends in AI. | Identify AI applications and discuss Generative AI and LLMs. | ISA, ESE |
| M2-LO1: Explain agents, environments, and rationality in AI. | Describe agent-environment interaction and rational behavior. | ISA, MSE, ESE |
| M2-LO2: Apply PEAS representation for intelligent systems. | Construct PEAS models for real-world AI systems. | ISA, MSE |
| M2-LO3: Analyze agent architectures and multi-agent interactions. | Compare agent types and evaluate negotiation/trust mechanisms. | MSE, ESE |
| M3-LO1: Apply uninformed search techniques to problem-solving. | Solve problems using BFS, DFS, and Iterative Deepening Search. | ISA, MSE, ESE |
| M3-LO2: Apply informed search techniques using heuristics. | Solve search problems using Greedy Best First Search and A*. | ISA, MSE, ESE |
| M3-LO3: Analyze adversarial and local search techniques. | Evaluate Minimax, Alpha-Beta Pruning, Hill Climbing, and Genetic Algorithms. | MSE, ESE |
| M4-LO1: Explain knowledge representation techniques in AI. | Describe knowledge representation using logical formalisms. | ISA, MSE, ESE |
| M4-LO2: Apply propositional and first-order logic. | Represent facts, rules, and relationships using logic. | ISA, MSE, ESE |
| M4-LO3: Apply reasoning under certainty and uncertainty. | Perform inference using chaining, unification, resolution, and belief networks. | MSE, ESE |
| M5-LO1: Explain planning concepts used in AI systems. | Describe state-space, hierarchical, and conditional planning. | ISA, MSE |
| M5-LO2: Analyze various planning approaches. | Compare planning techniques for different AI problems. | MSE, ESE |
| M5-LO3: Explain learning paradigms in intelligent systems. | Describe PAC learning, statistical learning, and learning theories. | ISA, ESE |
| M6-LO1: Analyze AI applications in diverse domains. | Evaluate AI-based solutions in healthcare, business, education, cybersecurity, etc. | ISA, ESE |
| M6-LO2: Evaluate ethical and societal implications of AI. | Analyze issues related to fairness, bias, privacy, accountability, and transparency. | ISA, ESE |
| M6-LO3: Propose responsible AI solutions for societal challenges. | Design AI-based solutions considering ethical and societal constraints. | ISA, ESE |